

Automation

072

Using technology to perform tasks with minimal human intervention

Automation uses technology and control systems to perform tasks with minimal human intervention, increasing efficiency and consistency across manufacturing, software, and services.

● DEFINITION

The use of technology to perform tasks with reduced human assistance, ranging from simple mechanical processes to complex software workflows.

● WHY IT MATTERS

Automation increases productivity, reduces errors, and frees human workers from repetitive tasks, playing a central role in modern manufacturing and business operations.

● WORKING PRINCIPLE



Automated systems use pre-programmed logic, sensors, and control mechanisms to execute repetitive or rule-based tasks consistently without requiring step-by-step human input for each instance.

● QUICK FACTS

- ▶ Modern automated factories can run production lines with minimal human presence on the floor
- ▶ Robotic process automation software can mimic human interactions with computer applications to automate office workflows

● KEY TERMS

- Process automation
- Control system
- RPA
- Workflow

● CORE CONCEPTS

- ▶ Process automation
- ▶ Control systems
- ▶ Robotic process automation (RPA)
- ▶ Workflow optimization

● APPLICATIONS

- ▶ Manufacturing assembly line automation
- ▶ Software business process automation
- ▶ Automated customer service (chatbots)
- ▶ Automated testing in software development

● ADVANTAGES

- ▶ Increases speed and consistency of repetitive tasks
- ▶ Reduces human error in routine processes
- ▶ Frees workers for higher-value tasks

CAREER OPPORTUNITIES

Automation engineer, Process improvement specialist, RPA developer, Systems integration engineer

RESEARCH AREAS

AI-driven intelligent automation, Robotic process automation, Human-automation collaboration

● REAL-LIFE EXAMPLES

- ▶ Automated car wash systems
- ▶ Chatbots handling customer inquiries
- ▶ Automated email marketing workflows

● LIMITATIONS

- ▶ Initial implementation can be costly
- ▶ Can lead to job displacement in certain roles
- ▶ Requires ongoing maintenance and updates

INDUSTRY APPLICATIONS

Manufacturing, Business process outsourcing, Software development, Customer service

FUTURE SCOPE

AI integration is expanding automation beyond simple rule-based tasks into more complex, judgment-based processes.

● CURRENT TRENDS

Growing adoption of 'intelligent automation' combining traditional automation with AI decision-making capabilities.

● SUMMARY

Automation uses technology to perform tasks with minimal human involvement, boosting efficiency and consistency across industries.